EE/CprE/SE 491 WEEKLY REPORT 8

Apr 15, 2025* - *Apr 22, 2025

Group number: *16*

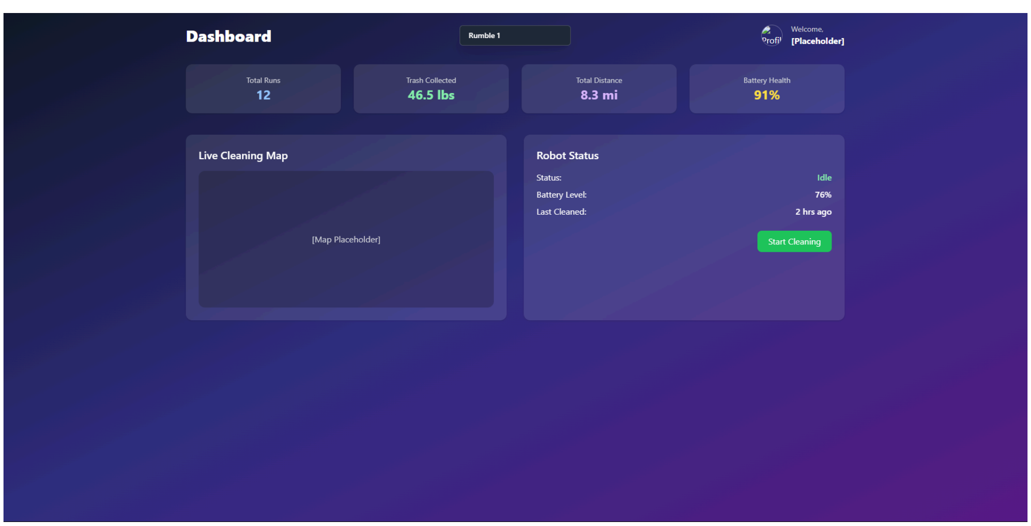
Project title: *Outdoor Roomba*

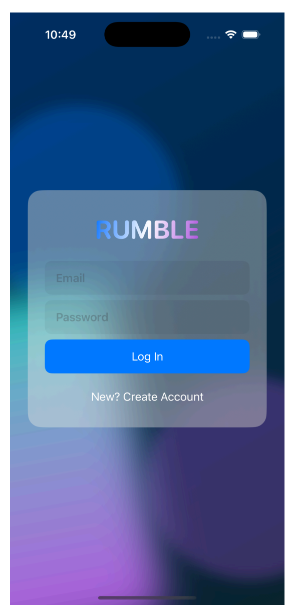
Client &/Advisor: *Prof. Diane Rover*

Team Members/Role:
Brodie M Bates - ***Embedded system***
Daniel Vergara Pinilla - ***Machine Learning***
Mitchell D Owen - ***Project Tester***
Om Shukla - ***Backend Developer***
Sudipta Halder - ***Simulation developer***

- **Weekly Summary**

This week there were concerns on which motors to use. We found that the dc motors can get damaged by stalling. So We have decided to hold on to motors from the ETG list and have decided to create a list of all the sensors, and other low priced components that will be used on the final design. We are planning to make a small prototype with cheaper motors and test out our sensors and object detection. One design change we have made is changing the radar A111-001-T&R we will use the radar A121-001-T&R which is the second iteration of the A111-001-T&R that has a higher range of detecting humans around 8 meters which is 4 time more then the first radar we picked. Also the app development is being worked on so far we have UI for the website and iphone app, as seen below(left os the website, right the is mobile app).





- **Past week accomplishments**

  - Sudipta Halder: laser cutted a basic platform for a four wheeled prototype, and attach wheels to it.

  - Brodie Bates: Gathering and researching materials for embedded collection of sensor data and motor controlling.

  - Daniel Pinilla: Worked on the login and sing up pages on the IOS app for the User interface

  - Om Shukla: Created website pages for the landing page, login, create account, and dashboard. These are written in basic HTML but will be converted to JS for the final product.

  - Mitchell Owen: Researching parts that will fit our desired specs like motors torque, amps, etc.

- **Pending issues**

  - Mitchell Owen: Running late on our progress. We don’t have anything bought yet to build our prototype.

- **Individual contributions**

NAME	**Individual Contributions**	**Hours this week**	**HOURS cumulative**
Brodie Bates	Research embedded materials for sensor data and motors.	4	39
Om Shukla	Relearned HTML to make websites.	10	37
Sudipta Halder	Built a small prototype, out of zip ties to test out the robot's sensors and object detection	5	44
Daniel Mauricio Vergara	Build the IOS app: login and sign up page	6	38
Mitchell Owen	Research specs for parts.	3	33

- **Plans for the upcoming week**

  - Sudipta Halder: finalize the sensors of the robot and build it.
  - Brodie Bates:Finish testing some potential embedded components. Assemble a testing breadboard with sensors and microcontrollers.
  - Om Shukla: Start converting websites from HTML to JS.
  - Mitchell Owen: Building what we plan on buying this week.
  - Daniel Mauricio Vergara: Work on the live tracking of the outside roomba being deplayed on the app

- **Summary of weekly advisor meeting**

In the advisor meeting, she told us to commit to a detailed design of an initial prototype. While exploring options and experimenting separately, we need to have a design in place, even if it changes later. Our prototype should be simple and still perform some basic functionality for use cases. Also we need to be clear, and especially explain these in a presentation. What requirements and specifications, what environment the robot will navigate, how it will pick up trash, what limitations, what mechanism to offload garbage. How is vision used, and have a solid terminology for our system.